INVESTIGATING SITE SURVEY PROCESS WITH PROTOCOL ANALYSIS AND AN EXTENDED FBS FRAMEWORK

YUE LUO[1], MANCHEN LIANG[2], LETONG GAO[3], YUCHUN ZHANG[4], CHENXI WANG[5], XIA SU[6] and WEIXIN HUANG[7]
[1,2,3,4,5,6,7]*School of Architecture, Tsinghua University*
[1,2,5]*{496835391|751969072|2576272411}@qq.com*
[3]*glt2012_zvv@126.com* [4]*zhang-yc17@mails.tsinghua.edu.cn*
[6]*su_xia1997@foxmail.com* [7]*huangwx@mail.tsinghua.edu.cn*

Abstract. In this paper, we try to investigate architectural site survey process by conducting experiment and quantitative analysis. 17 student volunteers were asked to practice site survey for a fixed design objective. With site survey process recorded along with sketching and utterance, we adopt protocol analysis and FBS ontology, which are widely used and discussed in design process research, as the basis of our analysis. Since site survey is a preliminary stage of architectural design, it differs from actual design process in many aspects. In this case, we extended the original FBS framework by adding two extra activities- Objective Processor and Subjective Processor- to better describe site survey process.

Keywords. Site Survey; Protocol Analysis; FBS Ontology; Architecture Education.

1. Introduction

Architecture students and professional architects conduct site survey as an important source of information and inspiration. However, there're very little research about the site survey, telling us what actually happens and how it affects following stages of design activity. With development in quantitative methods of evaluating design process, a natural idea is to conduct site survey experiments to yield results of analysis. In this paper, we introduce a quantitative research of architectural site survey based on experiment, protocol analysis and FBS ontology (J. Gero,1990). Additionally, since FBS ontology, which means function, behavior and structure, was initially put forward to analyze design activity, there is a lack of branches and connections for more specific design moves in site survey process. Our observation and understanding is that, during site survey process, designers process the features of site with their objective knowledge or subjective experiences. To help understand the design moves in the site survey process, we made extension to the FBS framework by adding two extra activities. With protocol analysis and the extended FBS framework we investigate our site survey experiment data and conduct multiple analysis.

RE: Anthropocene, Proceedings of the 25th International Conference of the Association for Computer-Aided Architectural Design Research in Asia (CAADRIA) 2020, Volume 2, 547-556. © 2020 and published by the Association for Computer-Aided Architectural Design Research in Asia (CAADRIA), Hong Kong.

2. Background

2.1. SITE SURVEY

Site survey is a preliminary activity for architectural design which inspects site where design is proposed. By conducting site survey, designer could look into site in detail and have immersive experience to find out potentials in actual design stage. Although enjoyed limited attention in traditional design process, site survey is considered to be a more critical step architect will take in the modern age. Having a more and more blurred ambit with the first draft, it is recognized as the fundamental activity for being aware of the basic disposition of the site, and also considered as the very first step where the design take place. Although site survey is a common and important practice among architects and architecture students, the process is rarely looked into. There are few related researches that inspects site survey process.

2.2. FBS ONTOLOGY

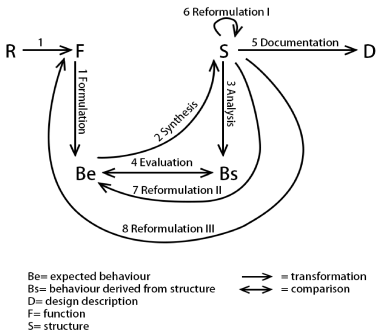

Figure 1. The FBS Ontology (J. Kan, J. Gero 2017).

FBS ontology (J. Gero,1990) is one of the most commonly used ontology in design process research which concludes design activities into 8 processes between 6 variables: F(function), Be (expected behavior), Bs (structure behavior), S(structure),R(requirement),D(documentation). Function (F) of design refers to its intention and purpose. Structure (S) refers to components and relationships of design. Behavior, including Be and Bs, refers to the performance, either the expected performance (Be), or the performance of Structure (Bs). By linking and transferring between these variables, FBS ontology could then describe design process with 8 processes, which is shown in Figure 1. The FBS ontology has been applied on many design fields and was shown effective as protocol analysis coding scheme.

3. Experiment Setting

In order to investigate the process of site survey, we adopt protocol analysis as our major research method. We set up a site survey experiment involving 17 volunteers and record the whole experiment process for further analysis. Considering the compactness between the site survey and the actual architectural design as well as the clear end of the experiment sign, subjects of our experiment are given a fixed design objective, which is locating a campus café in a certain area in a university campus. The site map is shown in Fig. 2. To investigate the possible correlation between architectural design experience and site survey activity pattern, we recruited 17 volunteers, among which 8 are architecture student ranging from senior to junior and 9 are from various other majors. All participants finished the site survey process independently in 10-15 minutes following "Thinking Aloud" principle.

Figure 2. Site map of experiment.

Experimental process:

1. The subjects are guided to the starting position (shown in the red dot in figure 2), and receive the architectural design assignment, site map, paper, pen and other materials and tools.

2. The subjects confirm the experimental content:site investigation for the design of coffee shop in this red dashed area. The objective is to select a specific location for the coffee shop and briefly explain the reasons for such selection: A. Observation and analysis of the objective environment B. subjective feelings in the venue and the reasons for these feelings. In the process of experiment, subjects need to reflect the thinking in the form of language (Think Aloud)

3. The subjects and the recorder go to the research site together to record the detailed image of the research process. The experiment can be ended after the subjects determine the location of the coffee shop and explain the reasons briefly. Over 15 minutes, the experiment will be ended, too.

4. After the experiment, the subjects return the materials and tools.

The entire experiment process is recorded, segmented and analyzed. Segmentation is conducted by separating design process into “design moves”, which are the smallest coherent operation detectable in design activity (Goldschmidt 1992). We then coded the design moves under an extended FBS framework of which the details are discussed in the next chapter.

4. Modifications and Adaptive Updates of FBS framework

Since site survey is a preliminary stage of design process, the focus and distribution of activity differs from the setting of original FBS ontology. Based on our observation from experiments, volunteers tend to focus mostly on the Requirement code, which include their thinking about both task document and site features. In this way, much of our concerned process would be unified under the R code if we proceed the analysis using the original FBS ontology. In this case, we propose additions and extensions to meet our demand.(Shown in Figure 3)

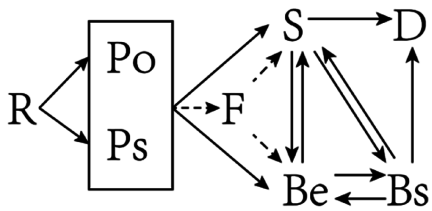

Figure 3. Diagram of the extended FBS framework.

We believe that the Requirement code here is uniform and not convenient for further classification, and the link between Requirement coding to Structure is not perfected. Between them, we added a coded box (Po and Ps codes in figure 3), which we call “processor”. “Processor” is the step where the analysis begins after the subjects have accepted the objective information from the outside world. They process the existing data in two modes of the processors - based on objective knowledge (Processor Objective) or based on subjective experience (Processor Subjective)-to derive different kinds of basic conclusions. Meanwhile in the original FBS coding, this part of the information is also classified as Requirement.

Table 1. Example of extended FBS codes.

FBS Code	**Content**
Requirement	So, I think we can check on the other side of the river directly.
Requirement	Um … let me see, this way is north, and that's south (looking at the picture).
Requirement	This way is west, and that's east.
Expected Behavior	So if … the café faces east rather than west
Structure	Maybe the west bank will be a better choice.
Requirement	Then … (looking at the picture) what are *Renjiantianshi* and *Fengwuqingtong*?
Requirement	I can see that there is a path along this side of the river.
Requirement	This way is … Zhishan Road.
Objective Processor	It is a main road of Tsinghua campus.
Requirement	Um … this way … I know this square.
Objective Processor	They have the ruin of the Summer Palace here.
Requirement	Here is the Lover Field.
Objective Processor	That's a brilliant work of Prof. Wang Lifang.
Structure Behavior	I think this place … is very suitable for constructing a café.
Subjective Processor	One fact is that it's relatively quiet.
Requirement	And it's also near to the Lover Field.
Requirement	Then there are some libraries next to it.
Requirement	But these trees are arranged in rows.
Structure Behavior	That's a little hard.
Structure Behavior	I think this piece of open land is nice.

In Table 1 we can see an example of segmented and coded site survey experiment. "It is a main road of Tsinghua campus" (line 9, row 2) is the conclusion that the subject personally derived from the objective condition (R code) "This side is the Zhishan road". It is from the basic knowledge "The Zhishan road is a main road in Tsinghua campus", which is not the knowledge that the tester provided to the subject. So this sentence is categorized not as the Requirement code, but as the conclusion derived from the objective knowledge processor (objective processor), coded as Po.

And "One is that it's relatively quiet." (line 15, row 2) is classified to the Ps code (subjective processor). Although its predecessor in the context is "I think this place ... is very suitable for constructing a cafe." (line 14, row 2), coded

as Bs, is a seemingly summative judgment of the design behavior coded as S (line 5, row 1)in the previous part of article, we actually think that the R coding and Ps coding the subject added (line 16-19, row 1) are the predecessors of the logical chain of the subject. They play the same explanatory role as the R and Ps codes while in the positive sequence. "Quiet" as the subjective feeling of the subject is classified to be the conclusion derived from "analysis based on subjective experience", which belongs to the new analysis code we have invented, namely Ps coding. Figure 4 shows the stacked bar graphs of a recorded and segmented site survey process with sliding window sized 14, showing the distribution of different design moves. The left graph adopts the original FBS ontology as coding scheme while the right graph adopts our extended FBS ontology. These graphs visually show how our extended framework differs from and outperform the original one. With the extended ontology, site survey process is better analyzed.

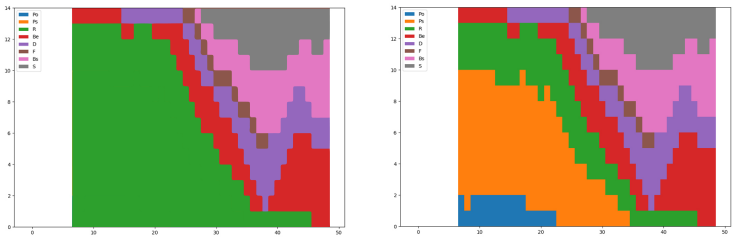

Figure 4. Comparation of previous and current process diagrams.

5. Analysis Results

Multiple analysis was conducted based on the the preprocessed data, including statistical analysis, process visualization, Markov chain analysis (J. Kan, J. Gero 2009), information entropy measurement (J. Kan, J. Gero 2018), Key Move, Ineffective Move, etc.

5.1. FBS ISSUE DISTRIBUTION

The participants' process visualization of FBS issue distribution is generated with sliding windows. The diagrams (shown in Fig. 5) show that all the 8 codes in the extended FBS on the timeline, indicating that the design moves in site survey process deal most often with Requirement(R), then objective analysis (Po) or subjective analysis (Ps) derived from requirement observation. Additionally, we notice how diverse peaks appear to be generated by the design moves that focus on Requirement(R), and further strengthened by design moves of 'Processer'(P). Compared with Students who have not received architectural education, architectural students have more obvious peaks and troughs in the dynamic issues diagram, which means they tend to concentrate on a single kind of design behavior over a period of time.

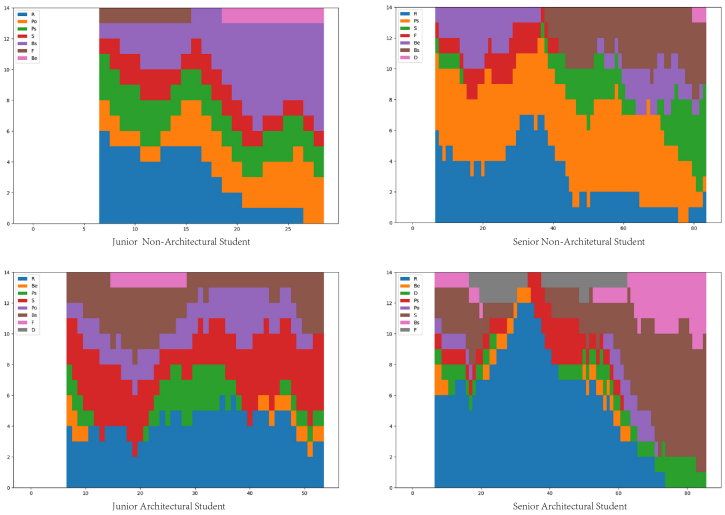

Figure 5. Example diagrams of process visualization.

When looking specifically at the design moves that focus on only external information collection(R) and processing of them (Po & Ps), we can find that some tend to collect external information, then perform specific design behavior(S) and make certain evaluations and analyses (Be & Bs). While others prefer to spread these two codes(R&P) throughout the experimental period. Through the analysis above, it can be concluded that there are two different types of information collection in site survey- centralized and distributed collection. Although non-architectural students also showed similar differentiation tendencies, the difference in the appearance of peaks and the amplitude of a single peak is more pronounced among architectural students.(Shown in Fig.6) From this observation, one can conclude that the designing mode of non-architectural students is more divergent and more average through the whole site survey process.

Another analysis of FBS coding is based on the dynamic process of switching between different codes. The Dynamic Process shows the coherence and efficiency of different design behaviors of volunteers. As shown in Fig. 6, there are almost a few peaks of different sizes in the charts of the architecture students, indicating that they have a concentrated code conversion during the design process. When we further analyze the categories of conversions, we can see that architectural students have more obvious peaks of ‘R-P’ type during the experiment, indicating that they accept external information and process it. In the middle or late stages, a main ‘S-S’ peak appears in their chart, showing that they are working in a centralized and coherent design. In general, this peak can also be found in the corresponding entropy map. In addition to the number and amplitude of peaks, architecture students generally have more conversion types than non-architectural students, which is reflected in the more complex color blocks in their charts. In summary, it can be inferred that the positive impact of architectural education on design thinking efficiency, proficiency and flexibility.

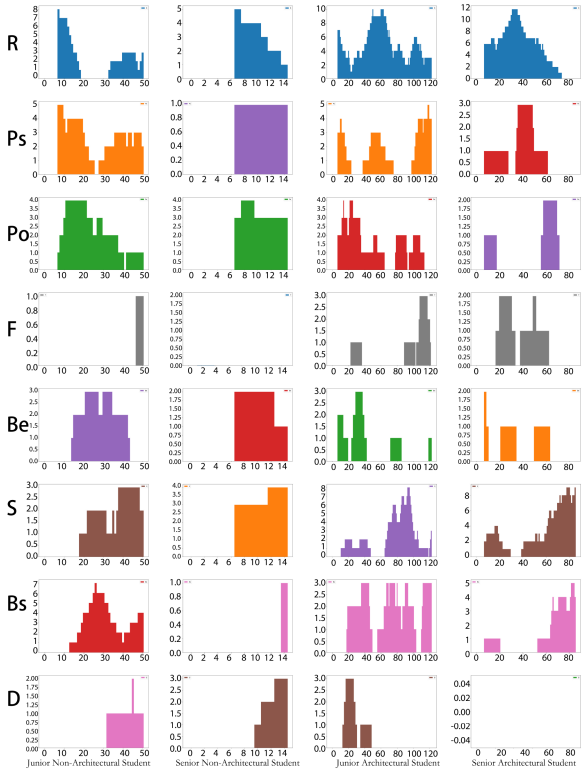

Figure 6. Example diagrams of dynamic issues.

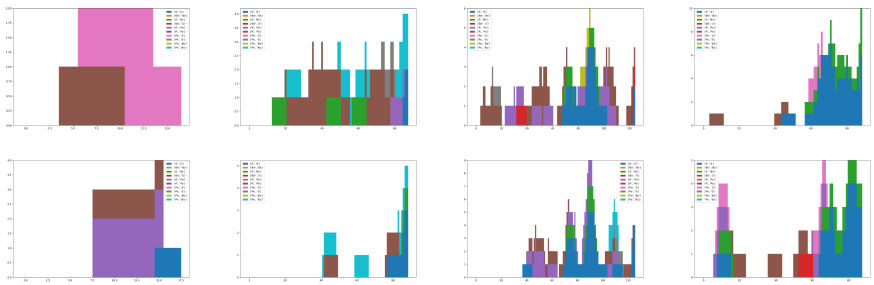

Figure 7. Example diagrams of Dynamic Process.

5.2. ENTROPY ANALYSIS

Information entropy is introduced by Shannon (1948) to measure the amount of information carried by messages. The related theory has dramatically influenced or even shaped today's information technology. In recent years, researchers have

transferred entropy into the field of design research. (Krus 2013, Gero 2011)... Specifically, Kan and Gero (J. Kan, J. Gero 2018) used it in analysis of design protocol to show the innovative potential of design processes. Here we also carried out entropy analysis using LINKODER (M. Pourmohamadi, J. Gero 2011), which is a protocol analysis software tool with FBS framework as basis. As shown in Table 1, the entropy value, which is positively related to the productivity of the design, of junior non-architecture students is lower than that of senior non-architecture students, also lower than that of junior architecture students. Moreover, the senior architecture students maintained high entropy values in the entire experiment. (Table 2.)

Table 2. Statistics of Average Entropy over the Linkograph.

Specialist subjects	Educational Level	Average Entropy		
		Forelink	Backlink	Horizontal link
Non-Architecture	Junior	1.632	1.743	1.234
	Senior	2.629	2.777	1.445
Architecture	Junior	2.132	2.333	1.263
	Senior	2.895	3.548	2.156

5.3. KEY MOVES AND INEFFECTIVE MOVES

A more detailed understanding can be found by looking at key moves and ineffective moves over the site survey process, using the linkograph visualization (shown in Fig.7). Key Moves can be distinguished using the number of backlinks and forelinks starting at specific design moves, showing that these more critical design moves actually correspond to design ideas or intentions. Positive correlation between the number of key moves and the level of architecture education is observed. Design moves that don't contribute to key moves and are only linked to neighboring moves are defined as Ineffective Moves. We find out that architecture student volunteers clearly have fewer invalid threads. This fact shows from another aspect that the volunteers who have received architectural education are more efficient in site survey process.

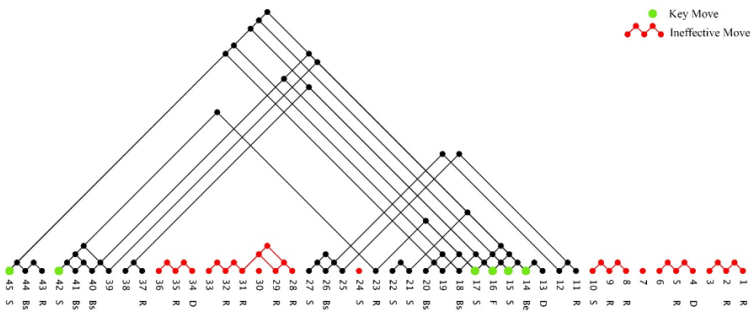

Figure 8. Example diagrams of Key Moves and Ineffective Move.

6. Conclusion

This paper presents analysis of site survey process based on experiment, protocol analysis and an extended FBS ontology. Our contribution is three-fold: Firstly, we novelly adopt quantitative analysis methods to investigate site survey process. Secondly, we extended the original FBS ontology to better fit site survey process. The extended ontology could possibly work on other preliminary design stages as well. Thirdly, we conduct multiple analysis based on experiment data and observe evident difference between two groups of volunteers, showing the effect of architecture education. From the analysis above, we can tell that in site survey process, architectural education provides a more efficient, proficient and flexible mind, followed with a more precise logic to get a designed result, which indicates that architectural education provides an enhanced capability in obtaining useful information from environment as well as in giving accurate descriptions.

References

Ericsson, K. A. and Simon, H. A.: 1984, *Protocol analysis: Verbal reports as data*, the MIT Press.

Gero, J. S.: 1990, Design prototypes: a knowledge representation schema for design, *AI magazine*, **11(4)**, 26-26.

Gero, J. S.: 2011, Fixation and commitment while designing and its measurement, *Journal of Creative Behavior*, **45(2)**, 108–115.

Goldschmidt, G. 1992, Criteria for design evaluation: a process-oriented paradigm, *in* Y.E. Kalay (ed.), *Evaluating and predicting design performance*, Wiley, Hoboken, 67–79.

Kan, J. W. and Gero, J. S.: 2017, *Quantitative methods for studying design protocols*, Dordrecht: Springer.

Kan, J. W. and Gero, J. S.: 2018, Characterizing innovative processes in design spaces through measuring the information entropy of empirical data from protocol studies, *AI EDAM*, **32(1)**, 32-43.

Kan, J. W. T. and Gero, J. S. 2009, Using the FBS ontology to capture semantic design information in design protocol studies, *in* J. McDonnell and P. Lloyd (eds.), *About Designing: Analysing Design Meetings*, Taylor & Francis, New York, 213-229.

Krus, P.: 2013, Information entropy in the design process, *Proc. ICoRD'13*, New Delhi, 101–112.

Pourmohamadi, M. and Gero, J. S. 2011, LINKOgrapher: An analysis tool to study design protocols based on FBS coding scheme, *in* S. Culley, B. Hicks, T. McAloone, T. Howard and Y. Reich (eds.), *Design Theory and Methodology*, Design Society, Glasgow, 2-294-303.

Shannon, C.E.: 1948, A mathematical theory of communication, *Bell System Technical Journal*, **27**, 397–423.